QUASI-COHERENCE OF THE DERIVED TENSOR PRODUCT AND ITS COMPUTATION ON AFFINES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- On affines the derived tensor product of the sheaves associated to module complexes is the sheaf associated to the derived tensor product of the module complexes. invoked at proof lines 637-638. Proves part (2).
- Being in D_QCoh is local on X. used at proof lines 639-642. Reduces part (1) to affines.
- On affines objects of D_QCoh are represented by complexes of modules. invoked at proof lines 643-644. Connects part (1) to part (2).

Source reference. Original lemma 08DV: sources/stacks-perfect/pinned-original-source.tex, source lines 544–544. Cited at proof lines 637-637 for the affine formula of part (2); the locality argument and conclusion are inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 643-644 to represent K and L by module complexes; the conclusion is inside the counted span.

Lemma 0.2. *Let X be a scheme.*

(1) *For objects K, L of $D_{QCoh}(\mathcal{O}_X)$ the derived tensor product $K \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ is in $D_{QCoh}(\mathcal{O}_X)$.*
(2) *If $X = \mathrm{Spec}(A)$ is affine then*

$$\widetilde{M^\bullet} \otimes^{\mathbf{L}}_{\mathcal{O}_X} \widetilde{K^\bullet} = \widetilde{M^\bullet \otimes^{\mathbf{L}}_A K^\bullet}$$

for any pair of complexes of A-modules $K^\bullet$, $M^\bullet$.

Proof. The equality of (2) follows immediately from Lemma 08DV and the construction of the derived tensor product. To see (1) let K, L be objects of $D_{QCoh}(\mathcal{O}_X)$. To check that $K \otimes^{\mathbf{L}} L$ is in $D_{QCoh}(\mathcal{O}_X)$ we may work locally on X, hence we may assume $X = \mathrm{Spec}(A)$ is affine. By Lemma 06Z0 we may represent K and L by complexes of A-modules. Then part (2) implies the result. □

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